EASA PPL(A) exam one-pagers

These are compact, topic-by-topic revision summaries for the nine EASA/Part-FCL PPL(A) theoretical-knowledge subjects, plus practical reference pages. The subject sheets follow the topics in the available Belgian course notes and question banks, explaining the main concepts, relationships, calculations, and distinctions rather than only giving general flight-safety reminders. They support study but do not replace the full course, current regulations, an operational checklist, or instructor guidance.

Print: Download all summaries as one PDF. Each topic starts on a new page; longer summaries span multiple pages.

The nine subjects

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How to use these notes

1. Learn the principle, then practise applying it to unfamiliar questions; do not memorize answers without understanding assumptions and units.
2. Check the current EASA rules and your competent authority's exam guidance for regulations and examination administration. National procedures, airspace, phraseology, and published minima can differ.
3. For an actual flight, use current NOTAMs/AIP, charts and weather products, the aircraft's current POH/AFM, and your instructor's approved procedures. Aircraft-specific limitations and approved data take precedence over generic examples here.

Course alignment and source hierarchy

Each subject sheet links to its corresponding detailed lesson notes and question bank in this repository. Use those longer materials for diagrams, worked examples, local classroom conventions, and full question practice.

Regulatory statements are linked to EASA's Easy Access Rules for Aircrew and Standardised European Rules of the Air (SERA). These are consolidated references; check the underlying current regulation and competent-authority material for operational use. The UK CAA's CAP 1298 is an older syllabus cross-check, not the controlling rule set for an EASA flight. Aviation fundamentals are cross-checked against official FAA handbooks, the UK Met Office, or NIST where suitable. Links are supplied on each sheet.

These notes do not reproduce protected question-bank content. Regulations and publications can change; sources were consulted in November 2026.

Air Law — course-topic summary

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Course scope: international and European rule structure; rules of the air; collision avoidance; aerodromes; airspace; altimetry; VFR minima; AIS; and local procedures. This follows the repository's Belgian course notes while keeping operational rules anchored to current EASA/state sources.

1. Who makes the rules?

Body or document	Role to remember
Chicago Convention / ICAO Annexes	International framework and SARPs; states implement and may notify differences
EU regulations	Directly applicable rules; **SERA**: rules of the air; **Part-FCL**: licensing
EASA AMC / GM	Acceptable compliance methods / explanatory guidance; not the same as the regulation
State authority / AIP / NOTAM	National implementation, local procedures and current changes
Belgium: DGLV/BCAA / Skeyes	Federal aviation authority / air navigation service provider; regions also have aerodrome responsibilities

2. Collision avoidance and aerodrome operations

Right-of-way is not permission to endanger another aircraft:

Encounter	Basic rule
Head-on	Both aircraft alter course to the right.
Converging	Aircraft with the other on its right gives way, subject to category-specific rules and exceptions.
Overtaking	Overtaking aircraft keeps clear and passes to the right.
Different aircraft categories	Towing aircraft, gliders, balloons and airships have specific priority relationships in SERA.

Circuit: departure/upwind → crosswind → downwind → base → final. Determine direction and height from local information.

PAPI indication	Position relative to glidepath
More red than white	Low
2 red / 2 white	On slope
More white than red	High

PAPI does not replace a stabilized approach or airspeed checks.

3. Airspace is three-dimensional

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POSITION → lateral boundary + base/top + altitude reference → class + active time
         → clearance / radio / transponder / flight plan → frequency
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CTRs start at the surface; CTAs/TMAs and airways can have stepped limits. Check NOTAMs for temporary restrictions. ATC clearance does not remove the pilot's safety responsibilities.

4. VFR weather minima and cruising levels

Airspace classes differ in access and ATS:

Class	VFR access and service (summary)
A	IFR only; VFR not permitted.
B	VFR permitted subject to clearance; all flights receive separation.
C	VFR requires clearance/radio; ATC separates VFR from IFR and provides traffic information between VFR flights.
D	VFR requires clearance/radio; separation is generally between IFR flights, with traffic information to VFR.
E	IFR is controlled; VFR generally needs no clearance, but is not separated from other VFR.
F/G	Advisory/information services; no ATC separation for VFR.

Services and class availability must be checked against the actual chart/AIP. VMC minima also depend on **altitude band, class, speed category, and state-specific rule**. The course examples use 5 km visibility below FL100 and 8 km at/above FL100; standard cloud separation is taught as 1500 m horizontal/1000 ft vertical, with special low-level and Class B wording in the bank. Always identify class and altitude first, then apply the current SERA/state table; “clear of cloud and surface in sight” is not interchangeable with standard separation.

The semi-circular cruising-level system is based on magnetic track/course. The course diagram uses 000°–179° eastbound and 180°–359° westbound, with odd/even thousand-foot sets respectively; VFR levels may add 500 ft where the applicable rule requires it. Confirm the relevant altitude, route direction and local exception rather than treating a classroom diagram as universal.

5. Altimetry and aeronautical information

Setting	Indication / reference
QNH	Elevation on the ground; altitude above mean sea level in flight
QFE	Height above selected aerodrome/reference datum
1013.25 hPa / QNE	Standard pressure reference for flight levels

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CLIMB:    QNH ── transition altitude ──> 1013.25 / flight level
DESCENT:  QNH <── transition level ──── 1013.25 / flight level
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Information	Purpose
AIP	Enduring information; local transition altitude/level and procedures
NOTAM	Time-critical changes; check validity and aerodrome status
AIRAC	Coordinated cycle for significant information changes
Runway condition report	Surface condition by runway thirds; use with aircraft performance data

Confirm chart edition and current QNH; “wet” alone does not determine braking performance.

Course notes: Air Law lesson notes · Air Law question bank

Primary references: EASA Easy Access Rules for Aircrew · EASA Easy Access Rules for SERA · Regulation (EU) No 923/2012

Human Performance — course-topic summary

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Course scope: the body at altitude, gas laws, vision/hearing/balance, health and fitness, information processing, stress, decision-making and human error. The central lesson is that normal human senses and judgement can fail in predictable ways; procedures and margins are safety defences.

1. Pressure, gases, and oxygen

The atmosphere is about 21% oxygen at altitude as well as at sea level, but total pressure and oxygen **partial pressure** fall with altitude. Less oxygen pressure is available to move oxygen from alveoli into blood. The ISA reference values are sea-level pressure 1013.25 hPa, temperature 15°C and density 1.225 kg/m³; they are a model, not the actual atmosphere.

Gas law	Relationship	Aviation connection
Boyle	Pressure × volume is constant at constant temperature	Trapped gas expands on climb: ear, sinus or gut discomfort
Charles	Volume varies with absolute temperature at constant pressure	Use kelvin when calculating
Dalton	Total pressure is the sum of partial pressures	Oxygen partial pressure falls with altitude
Henry	Gas dissolved in liquid depends on pressure	Pressure reduction after diving can release nitrogen bubbles (DCS)

Hypoxia is inadequate oxygen at tissue level:

Mechanism	What goes wrong	Example
Hypoxic / hypobaric	Insufficient oxygen pressure	Altitude
Anaemic	Reduced blood oxygen-carrying capacity	Carbon monoxide or blood loss
Stagnant / circulatory	Inadequate blood flow	Circulatory impairment
Histotoxic	Cells cannot use oxygen effectively	Certain toxins

Early effects can include impaired night vision, euphoria, poor judgement, fatigue, headache, reduced coordination and slower decisions. Cyanosis may occur later. The pilot may not recognize impairment. Use oxygen as required/available, descend promptly, and land/seek assistance as appropriate; applicable oxygen rules depend on operation and jurisdiction.

Hazard	Mechanism / cues	Response
Carbon monoxide	Odourless; binds haemoglobin. Headache, dizziness, nausea, confusion or fatigue; suspect cabin heat/exhaust	Stop suspected source; ventilate/use oxygen as directed; land promptly and seek medical help
Hyperventilation	Over-breathing lowers CO_2; tingling, dizziness, cramps, visual disturbance	If hypoxia is possible, address oxygen and altitude first; do not assume hyperventilation

Rapid pressure change can also cause ear/sinus barotrauma. A cold may block the Eustachian tube and prevent equalization, particularly on descent. DCS risk after diving depends on dive profile and timing: follow current diving/aviation medical guidance, not old textbook waiting-time examples.

2. Vision, hearing, and balance

Vision	Role / pitfall	Defence
Cones / rods	Central detail and colour / dim light and peripheral detection	Scan; use slightly off-centre viewing for dim objects
Dark adaptation	Takes time; bright light disrupts it	Preserve night adaptation; polarized lenses may interfere with displays
False horizon	Sloping terrain or cloud resembles a horizon	Cross-check instruments
Runway / depth illusion	Width, slope, haze, water, snow or darkness distort height cues	Brief runway; fly a stabilized approach
Autokinesis	A single light appears to move	Cross-check visual references and instruments

The semicircular canals sense angular acceleration; the otolith organs sense linear acceleration/gravity. In poor visual conditions they can mislead:

Illusion	Trigger	False sensation
Leans	Correcting an unnoticed slow roll	A turn in the opposite direction
Coriolis	Head movement during a sustained turn	Tumbling/vertigo
Somatogravic	Acceleration or deceleration	Pitch-up or pitch-down respectively

Keep the head still during turns and trust a reliable instrument cross-check when bodily sensation conflicts. Noise can damage hearing and impair radio comprehension; protect hearing and use clear phraseology/readbacks.

3. Fitness, workload, and decisions

Use **I'M SAFE**—Illness, Medication, Stress, Alcohol, Fatigue, Emotion—as a preflight self-check. Medicines, alcohol, smoking/CO, dehydration, illness and poor sleep may impair judgement even when the pilot feels capable. Get qualified medical advice; don't treat an old "hours since alcohol/medication" rule as personal clearance.

Attention selects some cues and filters others; expectation and confirmation bias can make a pilot overlook evidence that contradicts an initial diagnosis. Working memory is limited. The primary course question bank uses **5 ± 2 items for about 10–20 seconds**; other course/supplementary material gives different estimates, so use the primary bank's figure for that exam item. Chunking, written notes, checklists, readbacks and repetition reduce omissions.

Threat	Effect / defence
Excessive stress	Attention narrows; memory, coordination and decisions degrade
Hazardous attitudes	Anti-authority, impulsivity, invulnerability, macho, resignation, complacency → pause and challenge with objective limits
Risky shift	A group may accept more risk than its individuals
Error chain	Latent conditions + active errors + failed defences → interrupt with monitoring, checklists, training and reporting

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SITUATIONAL AWARENESS: perceive cues → understand → anticipate
SURPRISE: fly → navigate → communicate → diagnose / checklist
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Identify threats early; preserve options. A diversion, go-around, delay or cancellation may be correct.

Course notes: Human Performance lesson notes · Human Performance question bank

Primary references: EASA Easy Access Rules for Aircrew · FAA Pilot's Handbook of Aeronautical Knowledge

Meteorology — course-topic summary

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Course scope: atmosphere and heat; temperature and pressure; circulation and wind; thermodynamics and stability; clouds and air masses; fronts; fog, precipitation, icing, thunderstorms and mountain weather; reports, charts and VFR planning.

1. Atmosphere, temperature, and pressure

Most weather occurs in the **troposphere**, where temperature generally decreases with height; the tropopause height varies with latitude and season. ISA is a reference: MSL 15°C, 1013.25 hPa, density 1.225 kg/m³, with a tropospheric lapse rate near 2°C/1000 ft (6.5°C/km). Distinguish the **environmental lapse rate** (actual surrounding air) from the **adiabatic rate** of a moving parcel.

Heat process	Effect
Radiation / conduction	Sun heats surface / surface warms adjacent air
Convection / advection	Vertical heat transfer / horizontal transport
Evaporation / condensation	Absorbs / releases latent heat; saturated rising air cools more slowly
Inversion	Temperature rises with height; may trap haze/moisture and create low-level wind shear

Surface type, cloud, wind, latitude, time and season affect temperature; inversions may be radiation, subsidence or frontal.

Pressure is the weight of the air column. Isobars show equal pressure: tight spacing means a stronger pressure gradient and generally stronger wind. A trough/ridge extends low/high pressure. Pressure falls with altitude non-linearly; the classroom 27–30 ft per hPa conversion is only a local approximation. **QNH** is reduced to MSL using ISA; **QFF** is reduced using actual conditions; **QFE** references a local datum; **QNE** is standard pressure. Cold air makes true altitude lower than indicated; low pressure also reduces true altitude for a given setting.

2. Wind, circulation, stability, and cloud

Pressure-gradient force starts wind from high toward low pressure; Coriolis deflects it right in the Northern Hemisphere and left in the Southern. Aloft, wind tends to follow isobars; at the surface, friction slows and turns it toward low pressure. Tight isobars imply stronger flow. Near-surface terrain, sea/land contrasts and heating modify wind: anabatic daytime upslope, katabatic night drainage, sea/land breezes, channelled winds, and foehn warming/drying downwind of mountains. Turbulence may be convective, mechanical/shear, orographic, clear-air, or wake-related.

Rising air / stability	Typical rate or outcome
Dry parcel	Cools approximately **3°C/1000 ft**
Saturated parcel	Cools approximately **1.5°C/1000 ft**; rate varies with conditions
Stable air	Resists displacement → layered cloud, haze, smoother flow
Unstable air	Continues rising → cumulus, showers, thermals, turbulence

Compare parcel cooling with the environmental lapse rate. **Dew point** is the saturation temperature on cooling; a small temperature/dew-point spread indicates near saturation, not an exact cloud base. Moisture can also be expressed as vapour pressure, mixing ratio or relative humidity.

Clouds form when air cools to saturation or gains moisture. Stable air tends to stratiform cloud; unstable air to cumuliform cloud. Recognize Ci/Cs/Cc (high), As/Ac (middle), St/Sc/Ns (low/layered), Cu and Cb (vertical/convection).

Typical feature	Warm front	Cold front
Lift	Gentle ascent over cold air	Cold air undercuts warm air more steeply
Cloud	High cloud thickens/lowers into layers	Cumuliform cloud, potentially CB
Weather	Widespread, prolonged precipitation and poor visibility	Narrower band of showers, gusts and turbulence

These are typical patterns, not a substitute for observing the actual front.

Air-mass characteristics begin at the source region (continental/maritime, polar/tropical) and change over the route. Warm moist air over a cold surface is cooled and stabilized; cold air over a warmer surface is heated from below and destabilized. A polar-front depression develops through wave, mature-front and occlusion stages; pressure tendency and wind shifts help locate passage. In a typical Northern Hemisphere surface low, winds circulate counter-clockwise and inward; around a high, clockwise and outward, modified by friction and local terrain.

3. Hazards and weather interpretation

Fog	Formation
Radiation	Clear night, moist ground and light wind favour cooling to saturation
Advection	Moist air moves over a colder surface

Fog	Formation
Orographic	Moist air rises over terrain and cools
Frontal	Rain evaporates into cooler air below a front
Steaming	Cold air moves over warmer water

Precipitation may be continuous/stratiform or showery/convective. METAR visibility is not the same as runway visual range or an airborne slant view.

Airframe icing requires supercooled water and cold aircraft surfaces: clear/glaze ice forms from larger drops; rime from smaller drops; mixed ice combines them. Ice distorts aerofoil shape, increases drag and can reduce control/stall margins. Carburettor icing is a separate induction hazard caused by venturi and fuel-vaporization cooling, possible in moist air over a broad temperature range. Remove contamination before flight; use only approved aircraft procedures.

Ice type	Typical formation	Appearance
Clear / glaze	Larger supercooled drops spread before freezing	Smooth or translucent
Rime	Smaller drops freeze rapidly, trapping air	Rough, white, opaque
Mixed	Both mechanisms together	Irregular mixture
Hoar frost	Water vapour deposits on a cold surface	White crystals

Thunderstorms/CB bring severe turbulence, icing, hail, lightning, heavy precipitation and wind shear. Mature cells contain both strong up- and downdraughts. Mountain waves, rotors and lee turbulence may occur downwind of ridges; lenticular cloud can indicate wave activity. Do not treat a radar/satellite image as a complete go/no-go decision.

For planning, combine surface/upper-air charts, synoptic fronts, METAR/TAF, low-level area forecasts (including GAFOR where provided), low-level significant-weather charts, SIGMET/warnings, radar/satellite and winds aloft. GAFOR expresses regional VFR difficulty using forecast visibility/ceiling; decode the issuing service's zones, time blocks and legend. Radar depicts precipitation echoes, not all cloud or turbulence; satellite gives cloud coverage/top information, not runway visibility. Compare issue/validity times and route altitude; check trend, alternates and escape options. Forecast products describe a bounded area/time, not guaranteed conditions along the whole route.

Course notes: Meteorology lesson notes · Meteorology question bank

Primary references: EASA Easy Access Rules for Aircrew · UK Met Office aviation resources · FAA Aviation Weather Handbook

Communications — course-topic summary

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Course scope: radio propagation and equipment; ICAO alphabet and number pronunciation; message structure and phraseology; readback; distress/urgency; transponders; communication failure; Q-codes; and broadcast weather information.

1. Radio fundamentals

Radio waves are electromagnetic waves; frequency and wavelength are related by `frequency = wave speed / wavelength`. Aviation voice uses VHF (118.000–136.975 MHz) with **AM** and simplex push-to-talk: only one station transmits at a time, so listen before pressing PTT. VHF is mainly **line of sight**; altitude and terrain affect range. If no reception, check the selected COM/audio panel, frequency, volume/squelch, headset, microphone/PTT, power/circuit breaker and range before assuming a failure.

2. Speaking clearly and understanding numbers

Use the ICAO spelling alphabet (Alfa, Bravo … Juliett … X-ray, Yankee, Zulu) and standard digit pronunciation. Say headings, bearings, wind directions and flight levels digit-by-digit; say altitudes in hundreds/thousands as prescribed; transmit time in UTC. Frequencies are spoken as shown by the applicable channel display/phraseology. Visibility may be transmitted in metres or kilometres according to the value and local procedure. Never guess a number hidden by a noisy transmission—request “SAY AGAIN.”

A concise call normally gives **station called, aircraft callsign, position/intent/request**. Use the full callsign initially; abbreviate only after the station does so. Speak at an even pace and pause to allow another station to respond. Do not use casual wording in place of standard phraseology.

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STATION CALLED → YOUR CALLSIGN → POSITION / ALTITUDE → REQUEST OR INTENTIONS
                          ↑
           listen before transmitting
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Include only details appropriate to the call and applicable local procedures.

3. Readback and phraseology

Read back clearances and safety-critical instructions as required, including runway, taxi/hold-short, route, level/altitude, heading, transponder code, frequency and altimeter setting. Include the callsign. A readback is not a paraphrase: repeat the value accurately and listen for correction.

Phrase	Meaning
ROGER	Received; not “yes” or “will comply”
WILCO	Received and will comply
AFFIRM / NEGATIVE	Yes / no
SAY AGAIN / STANDBY / UNABLE	Repeat / wait / cannot comply

Distinguish information from an instruction and a clearance. A frequency change alone is not permission to enter controlled airspace. If an instruction is unclear or unsafe, query it; do not infer missing words.

4. Distress, urgency, and failure

Call	Meaning
MAYDAY	Distress: grave and imminent danger; immediate assistance required
PAN PAN	Urgency: safety concern, but not yet distress

Make the call on the current frequency if possible; state callsign, nature of problem, intentions, position/level/heading and assistance needed. Fly the aircraft first; transmit when workload permits. Other stations should not obstruct a distress call.

Squawk	Meaning
7700	General emergency
7600	Radio communication failure
7500	Unlawful interference

Radio failure: fly safely → check equipment/frequency → try alternate radio/frequency → blind transmit or relay if appropriate → select prescribed code → follow aircraft checklist and published state/AIP procedure. Controlled-airspace actions depend on the applicable rule.

5. Q-codes and weather broadcasts

Code	Reference	Direction / indication
QDM / QDR	Magnetic	To / from station
QUJ / QTE	True	To / from station
QNH / QFE / QNE	Pressure setting	MSL altitude / height above datum / standard pressure

Relative bearing is measured from the aircraft nose; don't confuse it with magnetic course. **ATIS** broadcasts current aerodrome information; note its letter and report it on first contact where required. **VOLMET** broadcasts meteorological reports for aerodromes/areas. METAR/TAF abbreviations such as FEW, SCT, BKN and OVC describe cloud amounts.

Worked Q-code and variation conversions

Course notes: Communications lesson notes · Communication question bank

Primary references: EASA Easy Access Rules for SERA · UK CAA CAP 413 (use as a phraseology study aid; national procedures may differ)

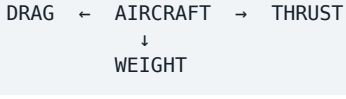
Principles of Flight — course-topic summary

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Course scope: airflow and aerofoil geometry; lift/drag; stability and controls; straight flight, climb/descent and turns; stall/spin; and propeller principles.

1. Aerofoil, airflow, and four forces

The four forces are **lift, weight, thrust and drag**. In steady level flight lift balances weight and thrust balances drag. The chord joins leading and trailing edges; the mean camber line describes aerofoil curvature. **Angle of attack (AoA)** is between chord and relative airflow; it is not pitch attitude or installed wing incidence. Relative airflow is opposite the flight path.



In steady level flight, the opposing forces balance.

An aerofoil's pressure distribution and momentum change produce lift. Bernoulli's speed/pressure relationship is useful, but a wing is not simply a Venturi tube and equal-transit-time reasoning is false. At the leading-edge stagnation point, local flow speed is near zero; the boundary layer develops along the surface. Separation grows as AoA approaches the critical value.

The basic equations are:

$L = \frac{1}{2} \rho V^2 S C_L \quad D = \frac{1}{2} \rho V^2 S C_D$

Dynamic pressure $q = \frac{1}{2}\rho V^2$. Since speed is squared, doubling speed multiplies dynamic pressure by four if other terms stay fixed. Lift coefficient depends on aerofoil, configuration and AoA. Increasing AoA initially raises lift; beyond the **critical AoA**, separation causes a sharp lift loss: a stall. A stall can occur at any attitude or airspeed if critical AoA is exceeded.

2. Drag, high-lift devices, and performance

Concept	Main effect
Induced drag	Lift-related; high at low speed/high lift; reduced by higher aspect ratio or ground effect
Parasite drag	Form, skin-friction and interference; rises strongly with speed
Total drag	Sum of induced and parasite drag; minimum between the two extremes
Maximum L/D	Best glide angle in still air; read the axes of a polar
Flaps	Increase camber (Fowler also increases area), lift and drag; change trim and can reduce climb performance
Slots/slats	Re-energize flow and delay separation, increasing usable AoA

Contamination changes the aerofoil and may cause earlier separation. Weight changes the lift required and stall speed, not the critical AoA itself. For the same configuration, accelerated stall speed rises approximately as $Vs \times \sqrt{\text{load factor}}$.

3. Controls, stability, and balance

Axis	Motion	Primary control
Longitudinal (nose-tail)	Roll	Ailerons
Lateral (wingtip-wingtip)	Pitch	Elevator
Vertical	Yaw	Rudder

Concept	Meaning / effect
Static / dynamic stability	Initial tendency after disturbance / subsequent motion and damping
Forward / aft CG	Generally more longitudinal stability and control force / less stability and potentially harder stall recovery
Ailerons / rudder	Roll may produce adverse yaw / rudder coordinates
Trim / mass balance	Relieves control force without moving CG / helps prevent flutter
Inclinometer ball	Slip/skid, not bank angle (“step on the ball”)

Tail surfaces and dihedral contribute to stability. Stay within the CG envelope, VNE and other aircraft limits.

4. Climb, descent, turns, stalls, and propellers

In a climb, thrust must overcome drag and the rearward component of weight along the flight path. **Vx** is best angle (height per horizontal distance); **Vy** is best rate (height per time). During descent, gravity contributes along the path; use the approved pitch/power technique.

In a coordinated level turn, `load factor n = 1/cos(bank)`. The horizontal lift component turns the aircraft; the vertical component must still support weight. More bank raises load and stall speed. Rate-one is 3°/s, but required bank depends on TAS. An uncoordinated stall can develop into autorotation.

Bank in coordinated level flight	Load factor	Stall speed relative to 1g
0°	1.00	1.00 ×
30°	≈ 1.15	≈ 1.07 ×
60°	2.00	≈ 1.41 ×

The stall-speed ratios assume unchanged configuration and the same 1g reference. Bank alone is not the load factor in a non-level or uncoordinated manoeuvre.

Condition	Wing state	Key distinction
Spin	Stalled, autorotating	Rotation with substantial height loss
Spiral dive	Unstalled	Steep descending turn with increasing speed and load

Prevent spins by maintaining coordination and avoiding the stall. Recovery is type-specific: use the POH/AFM and instructor, never generic steps as an operational procedure.

Propeller	Key point
Blade twist	Accommodates greater rotational speed at the tip
Fixed pitch	Optimized around a design condition
Constant speed	Changes blade pitch to maintain selected RPM, **not aircraft speed**
Yaw/roll tendencies	Torque, slipstream, gyroscopic effects and asymmetric thrust; coordinate with rudder

Course notes: Principles of Flight lesson notes · Principles of Flight question bank

Primary references: EASA Easy Access Rules for Aircrew · FAA Pilot's Handbook of Aeronautical Knowledge

Operational Procedures — course-topic summary

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Course scope: PIC responsibilities and documents; noise abatement; wind shear; wake turbulence; precautionary/emergency landings; ditching; fire/CO; contaminated runways; and survival/reporting topics.

1. Preparation and the pilot in command

Before flight	Check
Pilot / aircraft	Fitness, documents, airworthiness/defects, POH/AFM and checklist
Route / environment	Weather, NOTAMs, airspace, route and fuel
Loading / performance	Mass and balance; take-off and landing performance
People / cabin	Brief belts, exits, emergency actions and equipment; secure baggage

The PIC remains responsible for safe conduct, regardless of ATC clearance or passenger request.

Noise-abatement routes and profiles reduce disturbance but never override obstacle clearance, aircraft limitations or safety. Avoid unnecessary low flight over built-up areas; use published departure/arrival procedures and the approved aircraft profile.

2. Wind shear, microburst, and wake turbulence

Wind shear is a rapid change in wind speed and/or direction over a short distance:

Initial wind change on approach	Initial aircraft response
Headwind increases	IAS/lift may increase; path tends high
Headwind decreases or tailwind increases	IAS/lift may decrease; path tends low

Microburst sequence: increasing headwind → downdraft → decreasing headwind / tailwind. This is especially dangerous near the ground. Avoid storms and reported severe shear; if encountered, use the aircraft's approved escape procedure. Do not improvise flap/gear changes or a turn at low altitude.

Wake turbulence is the pair of vortices produced by a lifting wing. It is strongest behind a heavy, clean, slow aircraft. Vortices descend and drift with wind; a light crosswind may leave one over the runway. For departure behind a larger aircraft, aim to lift off before its rotation point and climb above its path where safe; for landing, remain above its approach path and touch down beyond its touchdown point. Follow ATC spacing and local procedures. Jet blast is a separate exhaust hazard on the ground.

Following a larger aircraft	Flight-path picture	Ground reference
Departing behind a departure	Climb above its path where safe	Rotate before its rotation point
Landing behind an arrival	Stay above its approach path	Touch down beyond its touchdown point

Published separation, ATC instructions, runway length and the aircraft's performance still govern.

3. Emergency and precautionary landings

Landing	When	Field assessment
Emergency	Circumstances force a landing	Size/shape, slope, surface/crops, obstacles and people
Precautionary	Chosen before a situation forces one	Wind, approach path, escape route and available height/time

Control the aircraft first; make calls and use the checklist when workload permits.

For ditching/over-water exposure, plan survival equipment for all occupants as required by the applicable rules and route. Brief exits and flotation. A life jacket inflated inside the cabin can obstruct egress; inflate it after leaving the aircraft unless the approved procedure directs otherwise. Aircraft-specific ditching speeds, flap, door and evacuation actions are from the POH/AFM.

4. Fire, smoke, and carbon monoxide

Fire response depends on the source. A suspected electrical fire may require electrical isolation; an engine/exhaust fire or cabin smoke may require fuel, mixture, heat/ventilation or engine actions in a different order. Use the checklist for the actual aircraft. Do not restart an engine that has been on fire unless the approved procedure specifically directs it. Water is generally unsuitable for electrical/fuel fires; use the installed extinguisher as trained, ventilate only as directed and land as soon as practicable.

Carbon monoxide is colourless/odourless and may enter through a failed heat exchanger. Headache, dizziness, nausea and confusion are warning signs. Stop the suspected source, ventilate/use oxygen if available as directed, land promptly and seek medical help.

5. Runway condition and performance

Water, slush, snow, ice, frost or other contaminants can increase rolling resistance, reduce braking and directional control, and increase take-off/landing distance. Standing water can cause aquaplaning: tyre contact and braking effectiveness fall, so stopping

distance increases. Wet grass also degrades performance. Read the runway condition report by runway third and use current aircraft performance data; a runway's “wet” label alone is not enough to compute distance.

Operational surface descriptions include **damp** (moisture changes the surface colour), **wet** (soaked but without significant standing-water patches) and water patches. The course/bank definition of a contaminated runway is more than 25% of the runway area being used covered by water, slush, snow or ice deeper than 3 mm; verify the current applicable definition and reporting rule. Learn runway-condition codes from the current source rather than inferring braking from a surface word.

6. Other course areas

Topic	Remember
Local operations	AIP circuit direction/height, runway markings/signs and noise-sensitive areas
Search and rescue	Flight plans support alerting; ELT and survival equipment aid recovery
Reporting / paperwork	Report occurrences when required; flight-time and document rules depend on operation

Course notes: Operational Procedures lesson notes · Operational Procedures question bank

Primary references: EASA Easy Access Rules for Aircrew · EASA Easy Access Rules for SERA. For aircraft emergencies and performance, use the current POH/AFM and checklist.

Flight Performance and Planning — course-topic summary

← Back to index

Course scope: mass/balance; fuel and flight-log planning; take-off, climb and landing performance; pressure/density altitude; wind components; chart/table reading; and ICAO flight-plan information.

1. Mass, moment, and centre of gravity

Use the aircraft-specific POH/AFM loading data. The **datum** is the reference from which each arm is measured; **moment = mass × arm**; total CG is `Σ moments / Σ mass`. Start from the current empty/basic mass and include occupants, baggage, fuel and other variable loads using the correct units and arms. Confirm both maximum mass and CG envelope—not just total mass.

Check loading for take-off and landing: fuel burn can change mass and CG. Forward CG generally increases stability/control force and may limit flare; aft CG reduces stability and may make recovery difficult. Secure baggage so it cannot shift. Do not use an example aircraft's arms or fuel density for another type.

Class C172 example:

Phase	Mass	Total moment	CG = moment / mass
Loaded	2451 lb	103,100 lb·in	42.1 in
After specified fuel burn	2202 lb	91,100 lb·in	41.4 in

These are example aircraft/data, not operating limits. Recompute each phase and check both mass and CG against the envelope.

2. Take-off, climb, landing, and density altitude

```
TAKE-OFF: ground roll → airborne segment → specified obstacle height
LANDING:  specified screen height → touchdown → landing ground roll
```

Chart input	Typical effect / check
Higher mass, pressure altitude or temperature	Generally reduces performance margin
Headwind / tailwind	Normally shortens / lengthens ground distance
Slope and surface	Check uphill/downhill and wet, grass, soft or contaminated corrections
Configuration and footnotes	Apply the actual chart's corrections and obstacle height; do not borrow another aircraft's factors

Ground roll and total distance over an obstacle are different answers; course examples commonly use 50 ft/15 m.

Pressure altitude is found by setting 1013.25 hPa and reading the altimeter (or applying the specified correction). **Density altitude** is pressure altitude corrected for non-standard temperature; high DA means lower density and typically less engine/propeller performance, less lift at a given TAS and poorer climb/take-off performance. Read charts exactly as published; interpolate only as permitted and apply each adjustment once.

```
Pressure altitude + temperature correction → density altitude
High / hot conditions → lower air density → generally longer take-off and poorer climb
```

Climb term	Measures	Useful for
Vx	Best angle: height gained per horizontal distance	Obstacle clearance
Vy	Best rate: height gained per time	Gaining altitude in least time
Gradient	Vertical / horizontal distance	Comparing height gained with distance

Do not confuse rate, gradient and angle. For a rough small-angle check, a 5% climb gradient is about 3°.

3. Fuel, distance, and flight logs

```
Distance + true course + wind + TAS → WCA / heading / GS → time / ETA → trip fuel
Trip fuel + taxi/start + climb + applicable alternate / contingency / final reserve
```

`time (h) = distance (NM) / GS (kt)`; use GS, not TAS. Verify usable fuel, grade, quantity, consumption units and the correct density when converting volume to mass.

Update the plan from actual groundspeed and weather at checkpoints. Recalculate remaining time and fuel; plan alternates and diversion decision points while options remain. A flight log commonly records checkpoint, frequency, minimum safe altitude, planned altitude/FL, track/heading, GS, distance, ETE/ETA/ATA and fuel.

4. Wind and weight/performance calculation discipline

Resolve wind relative to runway/track: headwind `W cos θ`, crosswind `W sin θ`. The crosswind limit is an aircraft/operator constraint; the maximum wind for a specified crosswind component is `component / sin θ`. Use exact trig or a flight computer when needed.

For weight/balance and charts:

1. Write values with units and use consistent mass/arm/moment units.
2. Verify datum, loading condition, chart title and configuration.
3. Check all chart notes and interpolation steps.
4. Compare result with limits and margins for every phase, including landing.

5. Flight plan and paperwork

Know the purpose and main information in an ICAO flight plan: aircraft identification/type/equipment, flight rules/type, departure and EOBT in UTC, speed/level/route, destination/total EET/alternates, other information, endurance, persons on board and survival/emergency equipment. Filing/activation/closing and when a flight plan is required depend on the state and operation; use current AIP/authority rules.

Course notes: Flight Performance and Planning lesson notes · question bank

Primary references: EASA Easy Access Rules for Aircrew · FAA Pilot's Handbook of Aeronautical Knowledge · NIST conversion factors. The current aircraft POH/AFM controls.

Aircraft General Knowledge — course-topic summary

← Back to index

Course scope: aircraft structure and materials; flight controls; piston engines and propellers; fuel, ignition and cooling; electrical/oil systems; and flight/engine instruments.

1. Structure, controls, and landing gear

Structure	Load path / function
Ribs / spars / stringers	Shape wing / major spanwise members / stiffen skin
Grid or space frame	Frame carries loads; covering is non-structural
Monocoque	Skin carries loads
Semi-monocoque	Stressed skin shares load with frames and stringers
Sandwich / honeycomb	Thin faces bonded to a light core

Main components: wing, fuselage, empennage, powerplant and landing gear. Composite damage can be hidden: use approved inspection methods.

Primary controls: ailerons (roll), elevator/stabilator (pitch), rudder (yaw). Flaps, slats, spoilers/speedbrakes and trim are secondary devices. Trim relieves control force; different tab types have different motions. Inspect control freedom, range, locking, hinges and surfaces as the checklist requires. Landing gear includes struts, fork/axle, wheel, tyre, brakes and (where fitted) shimmy damping. Inspect damage, tyre condition/pressure and leakage; bonding/electrical continuity matters during fuelling.

2. Piston engine, fuel, and ignition

Many light aeroplanes use a four-stroke opposed piston engine. Its cycle is **intake** → **compression** → **power** → **exhaust** over two crankshaft revolutions. Combustion should follow the intended spark; **detonation** is abnormal rapid combustion/knocking after ignition, while **pre-ignition** begins before the intended spark. High temperature, unsuitable fuel, excessive load/low RPM and mixture errors can contribute; use approved power/fuel settings.

Stroke	Piston	Main event
Intake	Down	Intake valve open; mixture enters
Compression	Up	Valves closed; mixture compressed
Power	Down	Combustion pushes piston down
Exhaust	Up	Exhaust valve open; gases leave

Dual magnetos are engine-driven ignition sources independent of the aircraft electrical system. Magneto checks normally produce a specified RPM drop; roughness can indicate plug/magneto trouble. A faulty grounding/P-lead circuit can leave a magneto live with the key off—treat the propeller as live. Follow the run-up and shutdown checklist.

Fuel system	Function / caution
Carburettor	Meters fuel into airflow; throttle controls airflow/power, mixture changes ratio, primer aids start
Carb ice	Venturi and evaporation cooling → ice in moist air, even without airframe ice; carb heat commonly reduces fixed-pitch RPM
Fuel injection	Meters fuel near cylinders; avoids venturi carb ice, not every induction fault
Fuel checks	Grade, selector, quantity, vents and drains; water collects at low points

Bond during refuelling and remove ignition sources.

3. Propeller, cooling, lubrication, electrical

A propeller is a rotating aerofoil; blade twist accommodates higher rotational speed toward the tip. Fixed pitch is optimized around a design range. A constant-speed propeller changes blade angle to maintain selected **RPM**, not aircraft speed. Know the controls, governor indications and operating limits for the installed engine/propeller.

Airflow/cooling fins and oil remove engine heat. Excess temperature can lead to detonation, wear or seizure; cowl flaps and mixture/power procedures vary by type. Oil lubricates, cools, cleans, seals and reduces noise; monitor pressure/temperature and use the POH service/check limits.

The electrical system may include battery, alternator/generator, regulator, buses, breakers, relays and loads. An alternator failure leaves only limited battery capacity; shed loads as directed. Magneto ignition may continue even if radios or electrically driven instruments fail. Identify which systems are engine-, vacuum-, pressure- or electrically driven.

```
ENGINE → magnetos → ignition
ENGINE → alternator → electrical bus → radios / electrical loads
                          ↑
                       battery
```

Schematic only: the electrical and ignition installations differ by aircraft; use its actual system diagram and checklist.

4. Flight instruments and indications

Instrument/system	Input or principle	Remember
ASI	Pitot/total and static pressure	Wind affects GS, not indicated airspeed
Altimeter	Static pressure	Setting determines altitude/height/FL reference
VSI	Static pressure change	Blockage/leakage causes errors; follow the POH
Direct-reading compass	Earth's magnetic field	Variation, deviation, dip, turning and acceleration errors
Gyro instruments	Rigidity and precession	Vacuum or electrical failure may remove different indications
Engine instruments	RPM, manifold/oil/fuel pressure, temperature and fuel sensors	Read ranges and warnings using the actual aircraft's AFM

IAS, CAS, EAS, TAS and GS are distinct. Read arcs and warning lights using that aircraft's legend; red/amber/green conventions are not a substitute for the AFM.

Course notes: Aircraft General Knowledge lesson notes · question bank

Primary references: EASA Easy Access Rules for Aircrew · FAA Pilot's Handbook of Aeronautical Knowledge. Use the current type-specific POH/AFM for all limits and procedures.

Navigation — course-topic summary

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Course scope: Earth coordinates and chart projections; distance/time/scale; wind triangle and heading; compass and bearings; radio aids; radar/transponder/GNSS; and integrated route, fuel, PET/ETP and holding calculations.

1. Earth, coordinates, charts, and scale

Concept	Key relationship
Latitude	N/S of equator; 1′ ≈ 1 NM, 1° ≈ 60 NM
Longitude	E/W of Greenwich; 1° ≈ `60 × cos(latitude)` NM
Earth rotation	15°/h; 1° longitude ≈ 4 min solar time; flight planning uses UTC
Great circle / rhumb line	Shortest spherical route / constant angle to meridians
Mercator / Lambert	Straight rhumb lines and scale increases with latitude / conformal conic suited to mid-latitudes
Scale `1:n`	Convert map and real distances to the same units before dividing

For short coordinate problems, combine latitude and longitude distances as perpendicular components. Handle N/S and E/W signs, including crossings of the equator. Check chart projection, validity, variation, obstacles, airspace and NOTAM/AIP.

2. Wind triangle, course/heading, and time

Distinguish **IAS/CAS/TAS/GS**: TAS is through the airmass; GS is over ground. Wind direction is reported **from**; reverse it to draw the vector. To maintain a desired track, apply wind correction toward the wind:

Speed	What it describes
IAS	Indicated by the airspeed indicator
CAS	IAS corrected for instrument and position error
TAS	Speed through the air
GS	Speed over the ground; use for leg time and ETA

1. Plot true course (TC) and wind vector.
2. Resolve head/tailwind and crosswind (or use an E6B).
3. Find wind-correction angle (WCA); `true heading = TC + WCA`.
4. Convert reference to magnetic/compass as needed and calculate GS.

Headwind reduces GS; tailwind increases it. `time (min) = 60 × distance (NM) / GS (kt)`; use GS for ETA, not TAS. Recompute from actual progress at checkpoints. For small errors, **1-in-60**: track error in degrees ≈ `60 × cross-track error / distance flown`; to regain the destination, add a closing angle based on distance remaining. Use only for small-angle approximations.

3. North references and radio bearings

True north is geographic; magnetic north follows Earth's field; compass north is the installed compass indication. **Variation** (charted, location-dependent) converts true↔magnetic; **deviation** (aircraft/equipment) is read from the compass card. Apply each correction once, with sign and convention stated, and normalize to 000°–359°. In the Northern Hemisphere, turning error uses UNOS (undershoot north, overshoot south); acceleration error on east/west headings uses ANDS.

Bearing	Magnetic	True
To station	QDM	QUJ
From station	QDR	QTE

Reciprocal bearings differ by 180°. RBI shows relative bearing from the aircraft nose; QDM is not the same as heading when wind/drift exists.

Worked course, heading, bearing and Q-code conversions: see navigation conversions.

Aid	What it provides	Important limitation
NDB/ADF	Relative bearing to LF/MF beacon	Night and shoreline effects can disturb bearings
VOR	VHF radial: magnetic bearing **from** station	CDI deflection is angular, not a fixed distance
DME	Slant-range distance to station	Difference from ground distance is greatest near overhead
VDF	Ground station derives bearing from VHF transmission	QDR locates aircraft from the station

For NDB/ADF, relative bearing (RB) is measured clockwise from the nose: `QDM = magnetic heading + RB`, normalized to 360°. Heading and inbound ground track still differ in wind. For VOR, set OBS, identify the station, and interpret CDI and TO/FROM;

two radials can define a fix. Check tuning, identification, mode and signal quality before relying on indications.

4. Radar, GNSS, and integrated planning

System	Provides / limitation
Primary radar	Detects reflected energy
SSR	Interrogates transponder for coded replies; know assigned squawk/ident and emergency codes
GNSS	Satellite position/time; depends on setup, database awareness and signal integrity

Cross-check chart, terrain and time; a display neither grants clearance nor replaces lookout.

For each route leg, calculate TC, WCA, true/magnetic heading, wind, GS, distance, ETE/ETA and fuel. Check terrain/minimum safe altitude, controlled/notified airspace, frequencies, alternates and NOTAMs. A **PET/ETP** is the point where the time to continue to the destination equals the time to return/divert under the stated wind/performance assumptions. For a simple out-and-back with constant TAS and opposite wind, distance from departure is `total route distance × (GS_out / (GS_out + GS_home))`; confirm which direction each groundspeed represents before substituting. Radius of turn and holding questions depend on speed, bank, wind and the specified pattern—do not mix a ground track with an air heading. In a standard-rate turn, 3°/s gives 360° in two minutes; required bank increases with TAS.

Course notes: Navigation lesson notes · Navigation question bank

Primary references: EASA Easy Access Rules for Aircrew · FAA Pilot's Handbook of Aeronautical Knowledge · NIST conversion factors. Use current official charts/AIP and a current approved flight computer for operational planning.

Units and conversions

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Distance, speed, and time

Quantity	Conversion
1 nautical mile (NM)	**1,852 m** = 1.852 km = 6,076.1 ft
1 statute mile	1.609 km = 0.869 NM
1 knot (kt)	1 NM/h = 1.852 km/h = 0.5144 m/s
1 km/h	0.540 kt
1 ft	0.3048 m
1 m	3.281 ft
1 degree of latitude	60 NM (approximately)
1 minute of latitude	1 NM (approximately)

Time: `distance (NM) ÷ groundspeed (kt) = hours`; fractional hours × 60 = minutes.

Decimal hours	0.1 h	0.25 h	0.5 h
Minutes	6	15	30

```
NM ── × 1.852 ──> km          ft ── × 0.3048 ──> m
NM <── ÷ 1.852 ── km          ft <── ÷ 0.3048 ── m
```

Example: 75 NM × 1.852 = **138.9 km**. Reverse the conversion by dividing by 1.852.

Mass, volume, and temperature

Quantity	Conversion
1 kg	2.205 lb
1 lb	0.4536 kg
1 litre	0.2642 US gal = 0.2200 Imperial gal
1 US gal	3.785 L
1 Imperial gal	4.546 L
Celsius to Fahrenheit	`°F = (°C × 9/5) + 32`
Fahrenheit to Celsius	`°C = (°F − 32) × 5/9`

Fuel volume is not fuel mass. Use the fuel grade's specified density or the aircraft's approved planning data when converting litres/gallons to kilograms/pounds. Do not treat US and Imperial gallons as interchangeable.

```
Volume (L) × fuel density (kg/L) = fuel mass (kg)
```

Example using a stated exercise density: 40 L × 0.72 kg/L = 28.8 kg. Use the density specified for the actual fuel and conditions.

Pressure and altitude

Quantity	Conversion / memory aid
1 inHg	33.864 hPa
1 hPa	0.02953 inHg
1 hPa pressure change	about 27 ft indicated-altitude change near sea level (rule of thumb only)
Standard pressure	1013.25 hPa = 29.92 inHg

Pressure-altitude correction is approximate and depends on the altimeter and atmospheric conditions. Follow the relevant altimeter-setting procedure; don't use the 27 ft rule as a terrain-clearance method.

Fast checks

- Write the **unit beside every value** before calculating. Cancel units as you would algebra.
- A groundspeed in knots gives distance in NM when time is in hours.
- Convert chart distance, fuel quantity, and aircraft mass to the units used by the POH/AFM before reading performance data.

- For a rough mental conversion, `kt × 2 ≈ km/h` (slightly high); use 1.852 for accurate calculations.
- Check whether a chart uses feet or metres, and whether a fuel figure is per hour, per engine, or total.

Sources: NIST Guide to the SI, Appendix B (conversion factors); NIST Appendix B.8; EASA Easy Access Rules for Aircrew (current aircraft-specific units and procedures: POH/AFM).

Core formulas and calculations

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The flight-planning triangle

Need	Formula
Time	`t (h) = distance (NM) / groundspeed (kt)`
Distance	`d (NM) = groundspeed (kt) × time (h)`
Fuel used	`fuel = consumption per hour × time (h)`
Fuel endurance	`usable fuel / consumption per hour` (before reserves/limitations)
Mass moment	`moment = mass × arm`
Centre of gravity	`CG = total moment / total mass`

Keep all inputs in the units required by the chart or aircraft data. For weight and balance, include every item, use the correct datum/arm, and confirm the resulting mass and CG fall within the approved envelope.

```
90 NM ÷ 120 kt = 0.75 h = 45 min
0.75 h × 24 L/h = 18 L trip fuel (excludes taxi, reserves and other allowances)
```

Aerodynamics and turns

Quantity	Formula / condition
Dynamic pressure	$q = \frac{1}{2} \rho V^2$
Lift / drag	$L = \frac{1}{2} \rho V^2 S C_L$ / $D = \frac{1}{2} \rho V^2 S C_D$
Load factor	`n = lift / weight`; `n = 1 / cos(bank)` in a coordinated level turn
Accelerated stall speed	$V_S(n) = V_S(1) \times \sqrt{n}$, same configuration and assumptions
Rate-one turn	`3°/s`; approximate radius in NM = `groundspeed (kt) / 188.5`

Use the specified standard-rate formula/table when required.

The lift equation is a model, not an operating limit. Use the AFM/POH for approved speeds, load limits, and performance.

Wind and performance

Calculation	Rule / meaning
Wind relative to track/runway	Headwind = `W × cos(angle)`; crosswind = `W × sin(angle)`
Approximate TAS from IAS	`TAS ≈ IAS × (1 + 0.02 × altitude in thousands of feet)`
Pressure altitude	Altimeter set to 1013.25 hPa, or approved pressure correction applied
Density altitude	Pressure altitude corrected for non-standard temperature

A headwind reduces groundspeed; a tailwind increases it. The TAS rule is rough: temperature and instrument/position errors matter. Hot/high/humid conditions reduce density and generally degrade take-off, climb, and engine/propeller performance.

For take-off and landing, interpolate charts only as instructed, apply all chart notes and corrections, and include surface, slope, wind, mass, temperature, pressure altitude, and obstacle effects as applicable. Never invent a correction factor.

Sanity checks

```
LABEL UNITS → ESTIMATE → CALCULATE → CHECK TREND / LIMITS → CHECK RESERVES
```

Use **GS**, not IAS or TAS, for map time/distance. Reserves are required fuel, not discretionary trip fuel.

Sources: EASA Easy Access Rules for Aircrew (PPL(A) syllabus and aircraft operational data); FAA Pilot's Handbook of Aeronautical Knowledge (principles, performance, and weight-and-balance fundamentals); NIST conversion factors. The aircraft's current POH/AFM is controlling for its actual performance.

VFR navigation calculations

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Route, track, heading

Term	Meaning / correction
Track / heading	Path over ground / direction the nose points; wind causes drift
Wind-correction angle (WCA)	Correct desired track toward wind; right positive, left negative
Variation	True ↔ magnetic; magnetic is less than true for east, greater for west
Deviation	Magnetic ↔ compass; follow the aircraft's deviation card

Label directions true, magnetic or compass; use the chart and card, not an unlabelled mnemonic.

```
TRUE TRACK ── + signed WCA ──> TRUE HEADING
TRUE HEADING ── variation ──> MAGNETIC HEADING ── deviation card ──> COMPASS HEADING
```

Example: track 040°T with a 7° left correction gives heading 033°T. With 4°E variation, magnetic heading is 029°M; apply the aircraft's deviation card separately.

Course, heading and bearing conversions: for the full true/magnetic/compass chain, QDR/QTE, variation, deviation and worked examples, see navigation conversions.

Wind triangle

For wind speed `W`, angle `θ` between wind direction and desired track, and true airspeed `TAS`:

Find	Formula
Crosswind	`W × sin θ`
Headwind	`W × cos θ`; negative for tailwind
Wind correction	`arcsin(crosswind / TAS)`, toward the wind
Groundspeed	`TAS × cos(WCA) − headwind`

These component formulas are a planning approximation; a flight computer or vector triangle is better for large angles or strong winds. Check whether a wind direction is reported **from** (meteorological convention) or **to** (vector direction).

Time, distance, and off-track

Find	Calculation
Time in minutes	`distance (NM) / GS (kt) × 60`
Distance in NM	`GS (kt) × time (min) / 60`
Track error, small angles	`60 × cross-track error / distance flown` degrees
Closing angle, small angles	`60 × off-track distance / distance remaining` degrees

To regain the destination, add the closing angle to the correction needed to maintain the original track, in the appropriate direction.

Worked 1-in-60: 5 NM right of track after 60 NM, with 60 NM remaining → track error ≈ 5° right, closing angle ≈ 5° → approximately **10° left** from the heading that produced the error, assuming unchanged wind and small angles.

Measure cross-track and along-track distances on the chart in the same units. The 1-in-60 rule is for small angles and short distances, not a substitute for replanning after a major deviation.

Chart and in-flight cross-checks

- Use charted magnetic variation, scale, legend, airspace vertical limits, frequencies, prohibited/restricted/danger areas, and terrain/obstacle data.
- Check position by more than one cue: time, heading/track, landmarks, terrain, and radio-navigation information if available. A single visual feature can be misleading.
- Recalculate ETA/ fuel and terrain clearance when wind or track differs from plan. Identify a safe diversion and decision point before conditions become limiting.
- Confirm route, altitude, QNH, airspace status, NOTAMs, and suitable alternates using current official information.

Sources: EASA Easy Access Rules for Aircrew (PPL(A) navigation syllabus); FAA Pilot's Handbook of Aeronautical Knowledge (navigation fundamentals); NIST conversion factors. Use current official aeronautical charts/AIP and the approved flight computer for operational planning.

Weather products and briefing

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Build a decision-focused picture

> `BIG PICTURE → ROUTE / TIME → HAZARDS → OBSERVATION vs FORECAST → OPTIONS`

Step	Check
Big picture	Synoptic chart, fronts, pressure, significant weather and upper winds
Route / time	Area forecast, aerodrome forecasts/observations, warnings and change groups
Hazards	CB, icing/freezing level, turbulence, waves, low cloud/fog, shear and crosswind
Compare / decide	Trends, discrepancies, deterioration, alternates and escape options

Products are time- and location-specific; refresh the briefing before departure.

Common coded products

Product	What it tells you	What it does not do
METAR	Observed aerodrome wind, visibility, present weather, cloud, temperature/dew point, pressure and trend where issued	It is not a route forecast or guarantee of conditions between stations
TAF	Forecast aerodrome conditions and expected changes during its validity	It is not a precise point forecast for the whole route
SIGMET	Significant en-route hazardous phenomena over a defined area/time	It is not a complete weather briefing
GAMET / low-level area forecast	Regional low-level conditions and hazards, where issued	It may not resolve local terrain effects
AIRMET / national warning	Additional hazard information where provided by the state	Naming/coverage and availability vary
ATIS / VOLMET	Broadcast aerodrome or en-route weather/operational information	It does not replace a full briefing or current NOTAM/AIP check

METAR/TAF decoding cues

Illustrative METAR: `METAR EBBR 121450Z 24012KT 9999 BKN020 18/12 Q1015`

Group	Interpretation
`EBBR 121450Z`	Station; day 12 at 14:50 UTC
`24012KT`	Wind from 240° true at 12 kt
`9999`	Visibility 10 km or more
`BKN020`	Broken cloud, base 2,000 ft above aerodrome
`18/12 Q1015`	Temperature 18°C, dew point 12°C, QNH 1015 hPa

This simplified example is not current weather. Reporting conventions and units vary by state.

Cue	Read it as
Wind	**From** the reported direction; check true/magnetic convention; `VRB` variable, `G` gust
Visibility / `CAVOK`	State-prescribed units; `CAVOK` is a defined coded condition, not simply “perfect”
`FEW` / `SCT` / `BKN` / `OVC`	Cloud amount; base normally in hundreds of feet above aerodrome
`CB` / `TCU`	Convective cloud
Temperature / dew point close	Nearer saturation; does not prove fog or cloud
`BECMG` / `TEMPO` / `PROB`	TAF change groups; read time window and probability

Go/no-go is about the whole route and return, including terrain, airspace, daylight, aircraft limits, pilot currency, alternates, and a margin for forecast error. Legal VMC minima are not a guarantee of safe VFR weather.

Sources: UK Met Office aviation meteorology resources; Met Office METARs and TAFs guide; EASA Easy Access Rules for Aircrew; EASA Easy Access Rules for SERA. Use the meteorological service and terminology applicable to the state and route.

Navigation conversions: course, heading, bearings and Q-codes

← Back to index · Navigation calculations

1. Course, heading and north references

Term	What it describes	Reference
Course / track	Desired route / actual path over the ground	True or magnetic north
Heading	Direction the aircraft nose points	True, magnetic or compass north
Bearing	Direction of an object or station from a position	True or magnetic north; relative bearing is from the nose
Variation (VAR)	Difference between true and magnetic north	Location and date; use chart
Deviation (DEV)	Error in the aircraft's magnetic compass	Aircraft, compass heading and equipment; use compass card

TC = desired true course; **TH / MH / CH** = true / magnetic / compass heading.

Signed correction	Positive (+)	Negative (−)
WCA	Right / clockwise	Left / anticlockwise
VAR or DEV	East	West

```
Desired TC ── + WCA ──> TH ── − VAR ──> MH ── − DEV ──> CH
Desired TC <── − WCA ── TH <── + VAR ── MH <── + DEV ── CH
```

Find	Formula using signed values	When used
Heading from course	`TH = TC + WCA`	Correct the desired ground track for wind
Course from heading	`TC = TH − WCA`	Recover the planned track using the same WCA
Magnetic from true	`MH = TH − VAR`; `MC = TC − VAR`	Change north reference, not direction
True from magnetic	`TH = MH + VAR`; `TC = MC + VAR`	Change north reference, not direction
Compass from magnetic	`CH = MH − DEV`	Obtain compass heading to steer
Magnetic from compass	`MH = CH + DEV`	Recover magnetic heading from compass
Find variation / deviation	`VAR = TH − MH`; `DEV = MH − CH`	Use comparable headings; express the difference as a signed angle

Check	Rule
Normalize directions	000°–359°; add/subtract 360° as needed
Signed difference across north	Use equivalent −180° to +180°: 003 − 358 = −355 = **+5°E**
Variation	Also converts true ↔ magnetic bearings
Deviation	Compass indication only; not a charted magnetic course or Q-code bearing
Compass card	Use deviation for the heading/equipment; do not reapply correction to a direct “steer” value
WCA	Depends on wind and desired track; heading alone cannot determine actual track

Worked heading conversions

Step	East variation, west deviation	West variation, east deviation
Given TC and WCA	TC 040°T; WCA 7° left (−7°)	TC 350°T; WCA 8° right (+8°)
True heading: TC + WCA	040 − 7 = **033°T**	350 + 8 = **358°T**
Magnetic heading: TH − VAR	033 − 4°E = **029°M**	358 − (−5°W) = 363 → **003°M**
Compass heading: MH − DEV	029 − (−2°W) = **031°C**	003 − 2°E = **001°C**

```
Reverse check: 031°C ── + (−2° DEV) ──> 029°M ── + 4° VAR ──> 033°T
               ── − (−7° WCA) ──> 040°T desired course
```

Here **°C means compass**, not degrees Celsius.

2. Q-codes: which end of the line, and which north?

Bearing	Magnetic north	True north	Meaning
To the station	**QDM**	**QUJ**	Direction from aircraft towards station

Bearing	Magnetic north	True north	Meaning
From the station	**QDR**	**QTE**	Direction from station towards aircraft

```
                FROM station: QDR (M) / QTE (T)
      [STATION] ──────────────────────────────────────> [AIRCRAFT]
      [STATION] <────────────────────────────────────── [AIRCRAFT]
                  TO station: QDM (M) / QUJ (T)
```

For local flat-chart exercises, TO ↔ FROM = **±180°** with the same north reference. QDR is magnetic FROM, like a VOR radial; QTE is true FROM.

3. Q-code conversion map

Take east variation as positive and west as negative:

```
            + signed VAR
QDR (M) ──────────────────> QTE (T)          FROM station
        <──────────────────
            − signed VAR

            + signed VAR
QDM (M) ──────────────────> QUJ (T)          TO station
        <──────────────────
            − signed VAR

QDR ↔ QDM: reciprocal ±180°    QTE ↔ QUJ: reciprocal ±180°
```

Conversion	East variation	West variation
Magnetic → true: QDR → QTE, QDM → QUJ	Add	Subtract
True → magnetic: QTE → QDR, QUJ → QDM	Subtract	Add
Reverse TO ↔ FROM, same reference	Add or subtract 180°	Add or subtract 180°

Normalize to **000°–359°**; `360°` = `000°`. Variation changes north reference; deviation is not used to convert a magnetic bearing to true.

4. Worked bearing conversions

Given	Steps	Answer
QDR 022°M, VAR 10°E	Magnetic → true: 022 + 10	**QTE 032°T**
QTE 229°T, VAR 10°W	True → magnetic: 229 + 10 = QDR 239; reciprocal 239 − 180	**QDR 239°M; QDM 059°M**
QDM 138°M, VAR 10°E	Magnetic → true: 138 + 10 = QUJ 148; reciprocal 148 + 180	**QUJ 148°T; QTE 328°T**
QDR 005°M, VAR 10°W	Magnetic → true: 005 − 10 = −5; add 360	**QTE 355°T**

```
MH 095° + relative bearing 355° = 450° → QDM 090°M
QDM 090°M ── +180° ──> QDR 270°M ── +6°E VAR ──> QTE 276°T
```

Relative bearing is clockwise from the nose: **do not apply variation to it**. This bearing relationship works in wind, but does not give the wind-corrected heading or ground track.

```
CH 031°C ── + (−2° DEV) ──> MH 029°M ── + 045° relative bearing ──> QDM 074°M
QDM 074°M ── +4°E VAR ──> QUJ 078°T
QDM 074°M ── +180° ──> QDR 254°M ── +4°E VAR ──> QTE 258°T
```

These are bearings at that position, not necessarily the heading to steer.

Relative-bearing conversion	Formula, normalized to 000°–359°
Magnetic bearing **to** station	`QDM = MH + relative bearing`
Relative bearing to station	`relative bearing = QDM − MH`
True bearing **to** station	`QUJ = TH + relative bearing`
True bearing **from** station	`QTE = QUJ ± 180°`

5. Exam method

```
LABEL: TO/FROM + TRUE/MAGNETIC
       ↓
REVERSE DIRECTION: ±180°     CHANGE NORTH REFERENCE: apply VAR
       ↓
NORMALIZE 000°–359° → label Q-code + reference → plot QTE FROM station on true chart
```

Either conversion order works for these local exercises; QDM must be reciprocated and converted before plotting from the station.

> **Scope:** Real-world plotting requires current charted variation, the appropriate station/reference and any chart or navigation-aid conventions. At long distances, meridian convergence and station-specific magnetic references can matter; use the chart/AIP and question assumptions. A bearing to a station is not necessarily the heading to steer in wind.

Course notes: Navigation lesson notes: north references and conversions · Q-codes and reciprocity · Navigation question bank

References: ICAO PANS-ABC, Doc 8400 · UK CAA CAP 413 · FAA Pilot's Handbook of Aeronautical Knowledge.